



VANIKA SHARMA

ENGINEERING PRODUCT DEVELOPMENT
ROBOTICS (GLOBAL MERIT SCHOLAR)



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A fiercely motivated and results-focused individual, anchored by a solid academic foundation and a track record of thriving in demanding environments, consistently surpassing expectations. My keen analytical and problem-solving abilities are harmonized with a robust work ethic and a proactive stance toward addressing challenges. Highly organized and efficient, I excel in juggling multiple projects concurrently, prioritizing adeptly, and meeting deadlines with unwavering precision. I am enthusiastic about channeling my skills and passion into initiatives that nurture innovation, collaboration, and professional growth.

WORK EXPERIENCE

Lionsbot International (Griffin Labs): Robotics RnD Intern

October 2025- December 2025

- **Rapid Prototyping:** Refined skills in SolidWorks and Fusion 360 to design and iterate on robotic end-effectors and structural mounts.
- **Startup Agility:** Learned to operate within a lean, "founder-mindset" framework, prioritizing iterative debugging and fail-fast methodologies.
- **Impact:** Bridged the gap between laboratory simulation and real-world deployment by iterating on the physical design of the robot to accommodate complex terrain, directly influencing the final hardware architecture

RoboSub Competition : Hardware Member, Supervised by SUTD Organisation of Autonomous Robotics

Made an Under Water Robot

- **Systems Architecture:** Managed the physical layout of internal electronics to optimize thermal dissipation and minimize signal interference between high-current motors and sensitive IMUs.
- **Tools:** SolidWorks, Fusion 360, Rapid Prototyping, CNC Machining, Laser Cutting.
- **Weight Optimization:** Reduced overall vehicle chassis weight by 15% through generative design and strategic use of lightweight composites, allowing for increased battery payload.

Undergraduate Research Opportunity Program in SUTD

Paper on AI in Computational Education that was presented in Kohli Conference

- **Data-Driven Pedagogy:** Optimized automated grading scripts to identify unique student coding patterns, helping educators distinguish between collaborative learning and academic dishonesty in high-volume STEM cohorts.
- **Python Algorithm Development:** Engineered a custom plagiarism detection engine using Python, implementing similarity algorithms to analyze structural and semantic code similarities.

PROJECTS

Built a Digi Camera from scratch

Independently architected a bespoke digital camera, executing the complete product cycle from PCB design and embedded firmware to a custom 3D-printed enclosure.

MakanMap: Dynamic Tactile Map

Engineered an inclusive HMI (Human-Machine Interface) for the visually impaired community, utilizing a dynamic haptic array to translate complex urban layouts into accessible, real-time tactile data.

Shelfi-mus Prime(Reconfigurable shelf)

Designed and Engineered an adaptive, sensor-driven shelving system that autonomously reconfigures its internal geometry to maximize storage density for high-volume, irregular parcel overflows

Loreal Hackathon: Discovery-folio

Redefined the luxury fragrance journey by architecting a collectible, tech-driven folio that utilizes body heat to trigger a controlled, time-released evolution of high-end scents.

Zygos: Awareness on Eating habits.

Architected a Python-based awareness game on PyCharm, leveraging real-time simulation logic to visualize the long-term health impact of dietary choices through an intuitive, interactive UX

Urban Sustainability Farms

Engineered a 'hidden-in-plain-sight' aeroponics farm powered by integrated windmills, merging biophilic design with renewable energy to turn underutilized building tops into productive green zones.

SKILLS

- **Design & CAD:** Fusion 360, SolidWorks, 3D parametric modeling, assembly design, basic FEA.
- **Electronics & Embedded:** Arduino, sensors & actuators, motor control, basic, PCB prototyping, Rapid prototyping
- **Programming & Robotics:** C/C++, Python, basic control systems, kinematics concepts, Arduino IDE.
- **Data Annotation & AI:** Image/video labeling, annotation QA, following detailed labeling guidelines, familiarity with ML datasets.
- **Engineering:** Problem solving, root-cause analysis, technical report writing, presenting design ideas, working in cross-disciplinary teams.